Journal of Liver Transplantation 18 (2025) 100260

Contents lists available at ScienceDirect

Journal of Liver Transplantation

journal homepage: www.elsevier.com/locate/liver

Original article

Automatic liver tumor classification using UNet70 a deep learning model

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ARTICLE INFO

Keywords:
Computed tomography
Deep learning
Unet70
Labelled images
3Dircadbs
Tumor (hepatocellular and metastatic)
No tumor

ABSTRACT

Diagnosing liver diseases using computed tomography (CT) images can be challenging even for experienced radiologists due to the complexities involved in evaluating the liver. Accurately determining the type, size and severity of tumors is often difficult. In recent years there has been a growing need for computer-assisted imaging techniques to aid in liver disease diagnosis ultimately improving clinical outcomes which in turn improves the life span of patients by early detection of the disease and treatment. This paper presents an innovative deep learning model UNet70 for liver tumor classification where CT images are categorized as either having a tumor (hepatocellular and Metastatic) or not. Our results show that the proposed model excels in terms of accuracy, sensitivity and dice score compared to other established algorithms and demonstrates excellent adaptability across various datasets. With an accuracy of 94.58 %, dice score of 94.73 % and sensitivity of 97.50 % the model outperforms existing methods showcasing its effectiveness.

1. Introduction

The liver one of the largest organs in the abdomen plays a vital role in bodily functions. When abnormal cells grow on the liver they form a lump called a liver tumor which is an indication of liver disease. Liver tumors can be classified as benign or malign. Benign tumors are non-threatening while malignant tumors are dangerous and if undetected in the early stages can lead to death. Malign tumors can signal serious liver conditions such as liver cancer, cirrhosis and hepatitis etc. [1]. Among this liver cancer has the highest mortality rate making early detection critical for timely treatment and improved patient outcomes. In 2020 9.05 million people were diagnosed with liver cancer with 8.3 million fatalities [2]. Projections indicate that the number of new cases could reach 40 million by 2040. Given this early detection of liver tumors is essential. However identifying liver tumors can be challenging due to the complexity of liver structure, its varying size, shape and the difficulty in manual analysis by radiologists. To address this it is crucial to develop automated techniques for tumor detection. Numerous methods both automatic and manual have been proposed to segment liver tumors from images particularly abdominal CT scans. Despite these advancements tumor segmentation remains a difficult task due to several factors including low contrast and blurred edges of CT images, the intricate structure of the liver, its size variability across individuals and the overlapping intensity of liver tissue with adjacent organs in CT scans.

Numerous automatic and semi-automatic techniques have been developed for the segmentation and classification of liver and its related diseases across various datasets [3]. These methods range from traditional approaches to advanced techniques utilizing machine learning and deep learning. Automatic segmentation techniques have demonstrated significant improvements often surpassing manual methods by reducing the need for expert intervention from radiologists [4]. Deep learning especially convolutional neural networks (CNNs) has been particularly effective for developing automated segmentation models as these techniques can quickly learn and adapt to variations, patterns and features in medical images [13] compared to traditional methods.

Contrast-enhanced computed tomography (CT) images are commonly used by medical professionals due to their cost-effectiveness and suitability for disease identification. Deep learning models provide valuable real-time feedback to doctors during surgeries and treatment, supporting decision-making throughout the process. The Fig. 1 below shows original contrast-enhanced CT images alongside their corresponding binary ground truths. In recent years fully convolutional neural networks have become widely used for medical image segmentation and classification tasks. Models like Alex Net, SegNet, UNet, Dense Net and ResNet have achieved significant improvements in metrics such as accuracy, sensitivity and Dice score enhancing both segmentation and classification results [5]. Although medical images can be

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https://doi.org/10.1016/j.liver.2025.100260
Received 26 December 2024; Received in revised form 21 January 2025; Accepted 30 January 2025
Available online 1 February 2025
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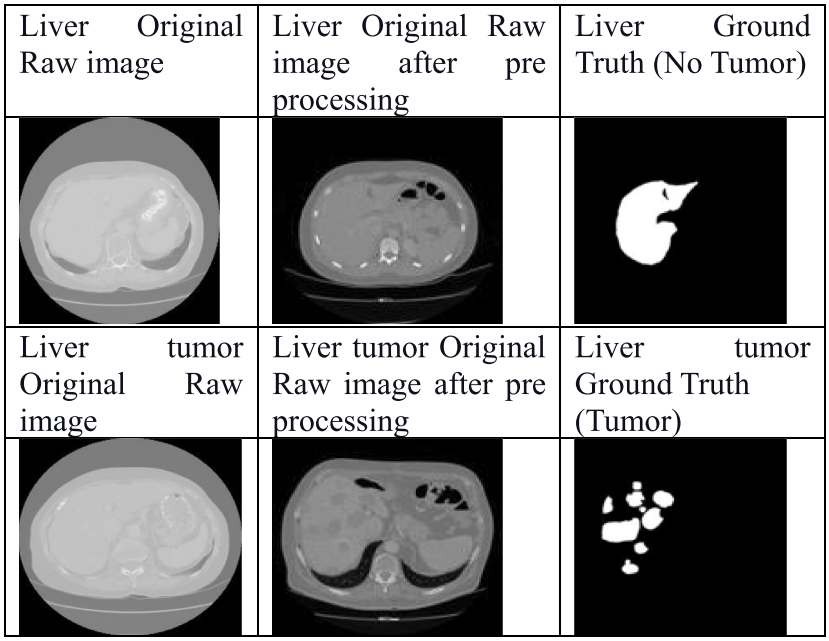

Fig. 1. CT images showing Liver and Liver tumor original and ground truth images.

acquired from various modalities such as MRI, CT, PET and X-ray, this paper focuses on classifying binary abdominal CT images into liver tumor and liver categories using a modified version of the UNet architecture (UNet70) applied to the 3Dircadb dataset. This paper focuses on classifying the liver and liver tumors using binary CT images of size 128 × 128. The proposed model outperforms existing methods in terms of key performance metrics including accuracy, sensitivity and Dice score demonstrating its effectiveness and robustness.

The main Highlights of the paper include

a. Developed an automatic deep learning model UNet70 which contains 70 layers for classification of the given data as tumor (Hepatocellular and Metastatic) and no tumor (liver).
b. Preparation of the 3Dircadb dataset a publicly available open-source resource, through preprocessing, cleaning and with the help of a radiologist.
c. Hyperparameter tuning using a trial and error approach to train the UNet70 model.
d. Development of a Confusion matrix for both training and testing dataset to evaluate the model performance by calculating different classification metrics.
e. Proposed model performed tremendously well by generating an accuracy of 94.6 %, dice score of 94.73 % and sensitivity of 97.5 % for testing dataset and similar results on training data itself.

This paper is structured as follows: Section 2 provides an overview of recent research on liver tumor classification. Section 3 outlines the proposed methodology. Sections 4 and 5 present the results and implementation, followed by the conclusion.

2. Literature survey

Segmentation is a major step in analysis of the medical images [19]. Deep learning techniques have been used to perform segmentation of various organs on different medical modalities. In this section different deep learning and machine learning techniques have been discussed which perform liver tumor segmentation and classification.

G Ignisha Rajathi [6] introduces a non-invasive method for classifying chronic liver disease using computed tomography (CT) images. The approach combines a hybrid feature selection technique using the whale optimization algorithm and simulated annealing to select 73 three-dimensional texture features grouped into seven categories. Classification is performed using an ensemble of support vector machine, k-nearest neighbour, and random forest classifiers to distinguish between five CLD categories: normal liver, fatty liver, metastasis, cirrhosis and cancer. The method was tested on clinical CT datasets, achieving an impressive 98 % accuracy, with sensitivity and specificity averages of 96 % and 93 %, respectively. These results significantly outperform existing methods, with an error rate as low as 1.9 %.

Amita Das [7] proposed a paper which introduces an automated framework for classifying liver tumors using CT images designed to improve early and accurate diagnosis. The framework called Multi-class Liver Tumor Identification involves four key steps: preprocessing the images, extracting features, optimizing the features using correlation-based selection and classifying the tumors with machine learning models. The study focuses on identifying benign tumors (hemangioma and cyst) and malignant tumors (hepatocellular carcinoma and metastasis). Out of 55 extracted features 20 were optimized for classification. Experiments with different Region of Interest sizes showed that Random Forest (RF) and Random Tree (RT) classifiers performed best, achieving accuracies of 97.48 % and 97.08 % on a 17 × 17 ROI size. These results outperform existing methods and demonstrate the framework's potential as a reliable diagnostic tool for radiologists.

Huawei Li [8] introduces a novel deep learning model MISNet for multi-class segmentation of CT images focusing on improving edge accuracy and reducing pixel misclassification. Utilizing a symmetric encoding-decoding structure with 3D Atrous Spatial Pyramid Pooling, MISNet effectively captures multi-scale features for precise organ segmentation. The model was tested on the COVID-19 CT Lung and Infection Segmentation dataset and the LiTS17 liver segmentation dataset achieving Dice Similarity Coefficients of 97.61 % for lungs and 98.78 % for liver segmentation outperforming state of the art models. MISNet uses a hybrid loss function (cross-entropy and Dice loss) to address sample imbalance and ensure stable training. The results highlight the model's superior accuracy and robustness, particularly for large organs, with clear 3D visualizations of segmentation outcomes.

Stephe S [9] proposed a paper with a novel approach for liver tumor detection and classification from CT scans using two models: Transformer-Based Attention Guided Network for segmentation and Vision Transformer-Gated Recurrent Unit for classification. The study

Table 1
Summary of pros and cons of different methods.

Author	Method and Dataset Used	Strengths	Limitations
G Ignisha Rajathi	KNN, SVM, RF & Clinical CT Images	Leverages clinical CT images for evaluating classification models	Limited sample size with only 73 CT images available
Amita Das	RF, RT & Clinical CT Images	Successfully differentiates between benign and malignant tumors	Small dataset size and restricted region of interest (17 × 17)
Huawei Li	MISNet & LITS17	Achieves high accuracy in organ segmentation (Dice score of 98.78 %)	Struggles with detecting small tumors in CT images
Stephe S	TAGN, VITGRU & LITS17	Outstanding results in classification metrics (accuracy, precision, recall)	Limited segmentation performance and small dataset (only 201 CT images)
Bheeshm Sharma	3D UNet, Patch-based Filtering & LITS17	Strong relevance to practical applications with enhanced accuracy	Faces issues with computational complexity and overfitting
Sumash Chandra Bandaru	Swingale & Duke Liver Dataset	Innovative approach for multi-class tumor classification	High computational demands and sensitivity to dataset quality
Srinivas Kolli	IPNN-BO & LITS17	Enhanced classification accuracy through Bayesian optimization for hyperparameter tuning	Restricted by a small dataset and assumptions in model design

utilized the LiTS2017 dataset comprising 201 abdominal-enhanced CT scans with preprocessing and augmentation techniques applied to improve data quality. TAGN enhances segmentation by integrating self-aware attention and Transformer self-attention modules within a UNet architecture, leveraging multi-scale skip connections to capture semantic features across scales. For classification VITGRU combines the Vision Transformer's hierarchical feature extraction with the GRU's temporal analysis for robust predictions. The TAGN model achieved an average accuracy of 84.65 % in segmentation while the VITGRU classifier outperformed baselines with an accuracy of 98.79 %, precision of 96.12 %, and recall of 95.64 %. This framework shows potential for aiding radiologists and improving diagnostic accuracy.

Bheeshm Sharma [10] developed a hierarchical deep learning framework for liver tumor segmentation and classification in 3D CT scans. The approach involves two steps: first, a modified 3D UNet architecture performs liver segmentation using a weighted Exponential Logarithmic Dice loss achieving a Dice score of 0.98 surpassing state of the art benchmarks. Second, a novel patch-based filtering technique extracts meaningful 3D patches from segmented liver regions. These patches are utilized by simpler models for tumor classification and segmentation achieving 89.1 % classification accuracy and a Dice score of 0.747 for tumor segmentation. The methodology tested on the LiTS17 dataset, highlights the efficacy of smaller networks combined with effective preprocessing techniques, demonstrating comparable performance to more complex models while emphasizing computational efficiency.

Sumash Chandra Bandaru [11] presented a paper which introduces Swingale, a novel framework for liver tumor classification using the Duke Liver Dataset. The study addresses challenges in medical imaging by incorporating state-of-the-art technologies like Swin Transformers, Generative Adversarial Networks, and attention mechanisms such as Convolutional Block Attention Module and Efficient Channel Attention. The GANs augment the dataset with synthetic images, while Swin Transformers capture complex spatial patterns. CBAM and ECA enhance feature extraction by emphasizing critical regions in the data. The proposed model, combining VGG19 with attention mechanisms and feature fusion, achieves an impressive classification accuracy of 99.29 %. This research sets a benchmark for multi-class liver tumor classification, improving diagnostic precision and demonstrating the potential of advanced deep learning methodologies in medical imaging.

Srinivas Kolli [12] introduced an Improved Probabilistic Neural Network with Bayesian Optimization for liver tumor segmentation and classification using the LiTS17 dataset. It addresses challenges in tumor detection such as overlapping tissue intensities and variability in liver structures by employing advanced preprocessing, segmentation and classification techniques. Preprocessing involved noise reduction, edge sharpening and image normalization followed by segmentation using customized masks for liver and tumor regions. The IPNN-BO model optimizes hyperparameters with Bayesian optimization to improve performance achieving superior results compared to traditional models

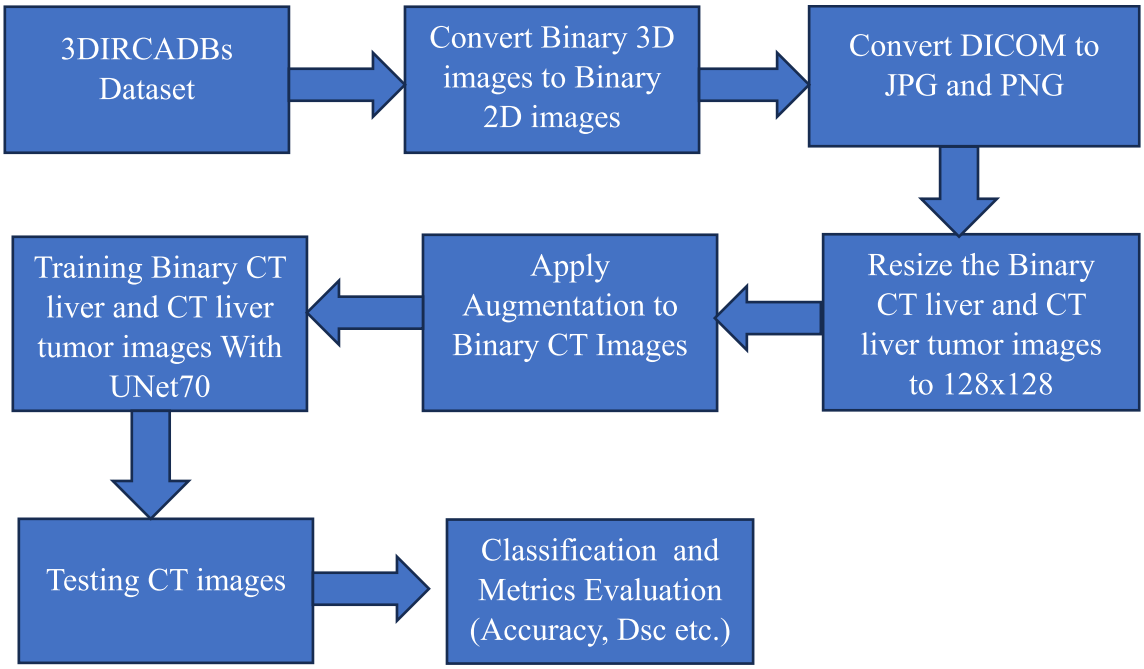

Fig. 2. Framework of the proposed methodology.

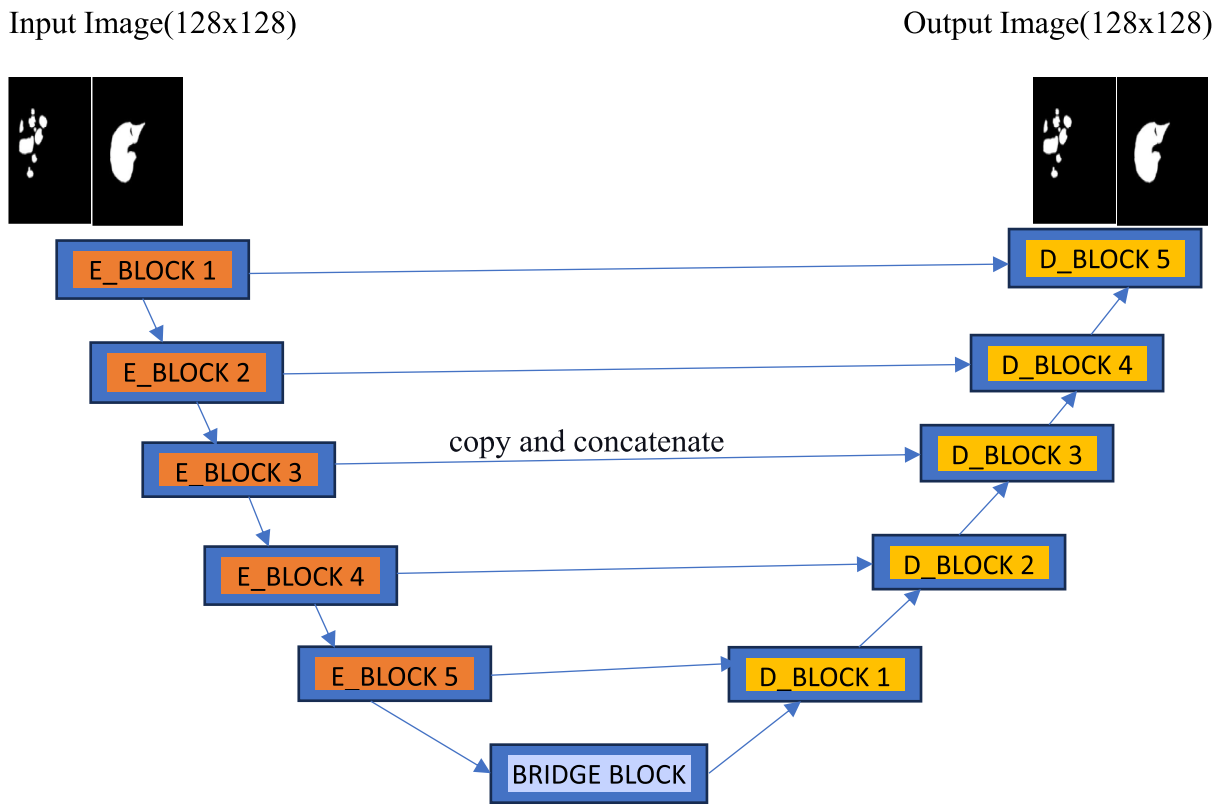

Fig. 3. UNet70 architecture.

like KNN, CNN and DCNN. The proposed method attained a segmentation Dice score of 0.897 and a classification accuracy of 99.25 %.

Table 1 displays the advantages and drawbacks of the different Methods discussed in the literature survey above.

The strengths and weaknesses of the proposed model (UNet70) compared to those of the models discussed in the literature survey are presented below.

Positives of the Model: The main strengths of our model lie in its ability to effectively classify liver tumors including both Hepatocellular Carcinoma (HCC) and Metastatic Carcinoma. The model achieved impressive results with an accuracy of 94.58 %, a Dice score of 94.73 % and a sensitivity of 97.5 %. Additionally, our approach benefited from the use of a large and diverse set of CT images for training and testing which contributed to the model's high performance. The diverse dataset preparation with the help of a radiologist.

Negatives of the Model: However, there are areas that require further improvement. The model still needs to be exposed to a broader range of datasets to enhance its generalizability. It is also necessary to test the model on real-time datasets to ensure its practical applicability in clinical settings. Furthermore, while the current results are promising there is still room for improvement in the model's metrics particularly in terms of dice score and Accuracy. Another challenge is that the model requires considerable time for training which could be optimized in future iterations.

3. Methodology

We established a deep learning model UNet70 to classify binary CT images of the liver and liver tumors from the 3Dircadb dataset. The methodology illustrated in Fig. 2 involves categorizing the binary CT images into liver and liver tumor classes. The dataset was utilized for both training and testing using the UNet70 model

The 3Dircadb [14] dataset was used to extract liver and tumor images from the available different organ datasets. The images originally in DICOM format were converted to JPEG and PNG formats with a resolution of 1024 × 1024. To align with hardware constraints the binary images were resized to 128 × 128. Data augmentation methods such as rotation and flipping were employed to increase the size of the training dataset. The processed data was utilized to train and test a custom UNet70 model, comprising 70 layers with its performance evaluated using metrics like true positives, true negatives, false positives and false negatives.

3.1. Dataset

The dataset for this research was sourced from the publicly available 3Dircadb dataset which includes 20 CT scans from different patients. From this collection, we specifically selected 3D liver CT scans and converted them into 2D images. In total 1760 binary CT images were extracted representing liver and liver tumor data and saved in JPG and PNG format. To accommodate computational limitations these images were resized to 128 × 128 pixels. For model training and evaluation, the dataset was split into 85 % for training and 15 % for testing [15].

3.2. Network architecture

The figure below illustrates the architecture of UNet70 network [16]. The designed model contains 5 blocks of encoder (down sampling) and 5 blocks of decoder (up sampling) parts along with one block of bridge which connects both encoder and decoder parts as shown in Fig. 3. The encoder consists of layers such as convolution, ReLU activation, max-pooling and dropout. The decoder includes up-convolution, up-ReLU, convolution, ReLU and depth concatenation layers. Meanwhile the bridge network integrates the encoder and decoder through bridge convolution, bridge ReLU, and bridge output layers.

Table 2
Hyperparameter settings.

Hyperparameter	Values Range
Input Image Size	128 × 128
Learning Rate	0.000125
Drop Factor	0.5
Drop Period	100
Epochs	200
Mini Batch Size	01
Optimiser	Adam
Learning Rate	Piecewise

4. Results and implementation

The proposed system for classifying liver and liver tumors is trained and evaluated using 1760 CT images which include both liver and liver tumors. The model employs binary segmented tumor data obtained from the UNet architecture applied to CT images from the 3Dircadb dataset. The UNet70 model which features various layers is used for feature extraction and the classification of liver tissue and liver tumors including Hepatocellular carcinoma and Metastatic carcinoma.

4.1. Evaluation metrics

During the verification phase various metrics such as accuracy, sensitivity, specificity, Precision and Dice similarity coefficient are employed to evaluate the performance of the deep learning model. The formulas for these metrics are outlined below along with their respective explanations.

(1) Accuracy [15]: It is the proportion of accurately identified pixels to the overall number of pixels in an image.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

(2) Sensitivity [15]: It is the percentage of true positive values that are correctly identified.

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{2}$$

(3) Specificity [15]: It is the percentage of true negative values that are correctly identified.

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{3}$$

(4) Precision [16]: It is the ratio of the model's positive predictions that are correct.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

(5) Dice score [17]: It is a measure of how similar the two samples are to each other.

$$\text{Dice} = \frac{2TP}{2TP + FP + FN} \tag{5}$$

Where TN= True Negative, TF= True Positive, FN= False Negative and FP= False Positive.

4.2. Implementation and testing

The proposed method was implemented using MATLAB 2024a [16] on a desktop computer. The hyperparameters [18] for training the UNet70 model were determined through multiple trial runs with the specific values listed in Table 2. The dataset was split randomly with 85 % used for training and 15 % for testing. The model incorporated a total of 70 layers and training was conducted for 200 epochs each consisting of 1520 iterations leading to a total of 304,000 iterations. The model took a total of 6528 mins and 32 ss to train on the provided dataset. The training results as shown in Figs. 4 and 5 demonstrate a gradual decrease in loss with increasing iterations which stabilizes after a certain point (Fig. 4). Likewise, the model's accuracy improves initially and then levels off as training progresses (Fig. 5).

4.3. Classification results

The proposed UNet70 model effectively classifies liver and liver_-tumor images (Hepatocellular and Metastatic carcinoma). The confusion matrices for both training and testing datasets are presented in Tables 3 and 4 respectively. Table 3 shows that the model correctly identified 693 liver images and 718 liver_tumor images out of a total of 1520 training images. In Table 4 the model achieved correct predictions for 117 liver images and 110 liver_tumor images out of 240 testing images. Figs. 6 and 7 illustrate the confusion matrices for training and testing dataset, where the model achieved an accuracy of 92.82 % for the training set and for the testing set it achieved an accuracy of 94.6 % with a sensitivity of 97.5 % and a dice score of 94.73 %. Fig. 8 provides a visual representation of the results obtained from the proposed model. Table 5 highlights the performance of the model using various evaluation metrics for both training and testing datasets. To validate the effectiveness of the proposed approach we compared its results with those of other classifiers summarized in Table 6. The comparison clearly demonstrates that the proposed model achieves superior accuracy, dice score and sensitivity on the testing data compared to alternative methods.

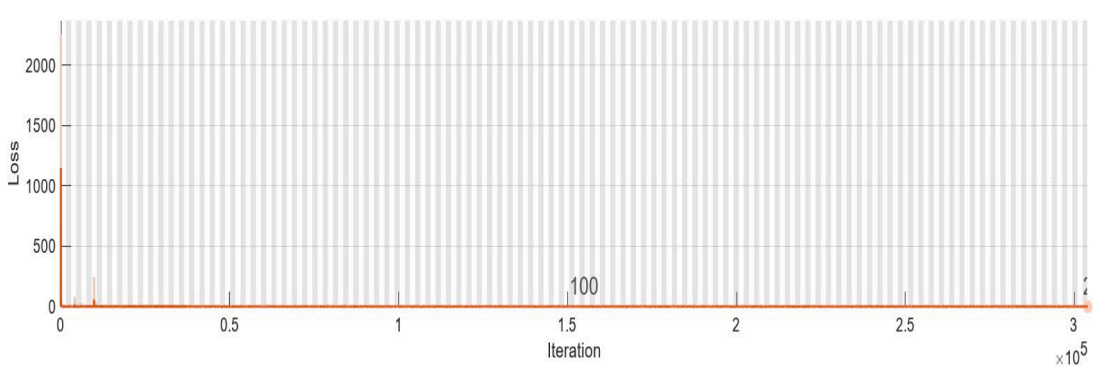

Fig. 4. Loss vs iterations.

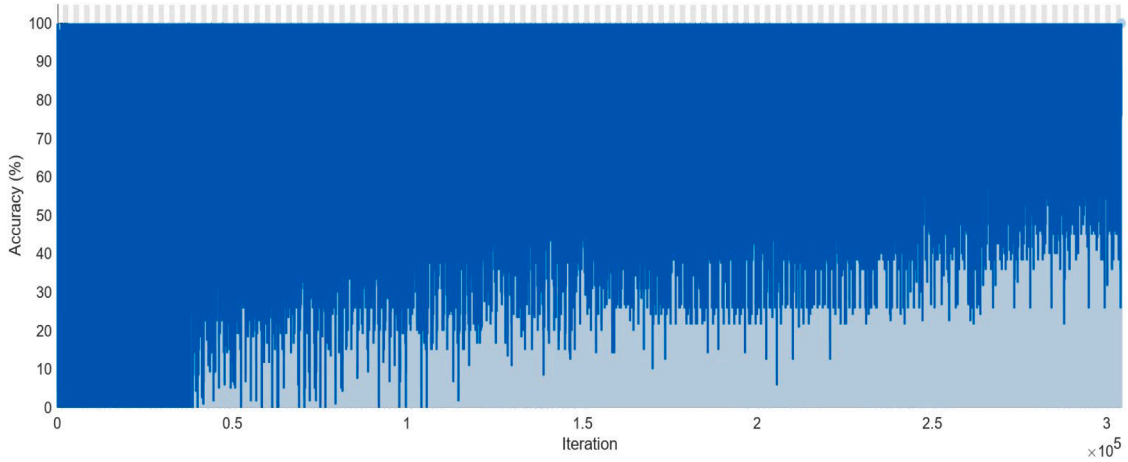

Fig. 5. Accuracy vs iterations.

Table 3
Training data results.

Training set		
Class	Liver	Liver_Tumor
Liver	693	42
Liver_Tumor	67	718

Table 4
Testing data results.

Testing set		
Class	Liver	Liver_Tumor
Liver	117	10
Liver_Tumor	03	110

Fig. 8 displays the outcomes from the UNet70 model applied to the testing dataset. The second column displays the ground truth labels for the tumor (Hepatocellular and Metastatic Carcinoma) and liver in a binary format. The third column presents the classification results for the liver and liver Tumor.

5. Conclusion

The proposed UNet70 model demonstrated excellent performance in accurately classifying both the liver and liver tumors. It achieved an impressive accuracy of 94.6 %, dice score of 94.73 % and a sensitivity of 97.5 % on the 3Dircadbs dataset and the model outperformed various deep learning and machine learning classifiers. It utilized a total of 31,000 learnable parameters while training on 1520 CT images highlighting the robustness of the model. In future work we aim to apply the model to the LITS (Liver Tumor Segmentation) dataset which includes a wider range of liver diseases and explore acquiring labelled datasets from multiple hospitals for further training and testing. Additionally, we plan to extend our research to include the segmentation and classification of other organs and their associated diseases facilitating more timely diagnosis. Finally, we intend to adapt the model to work with various medical imaging modalities such as MRI, T-MRI and PET scans.

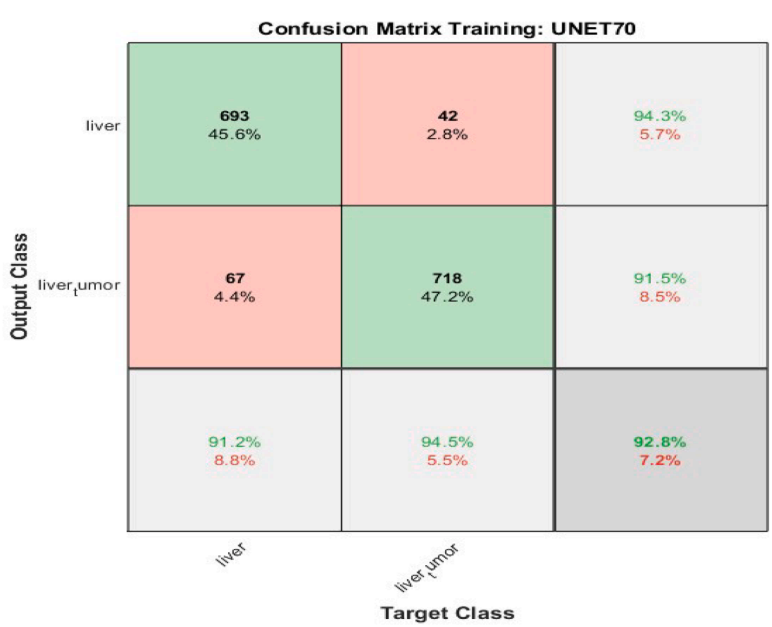

Fig. 6. Confusion matrix for training.

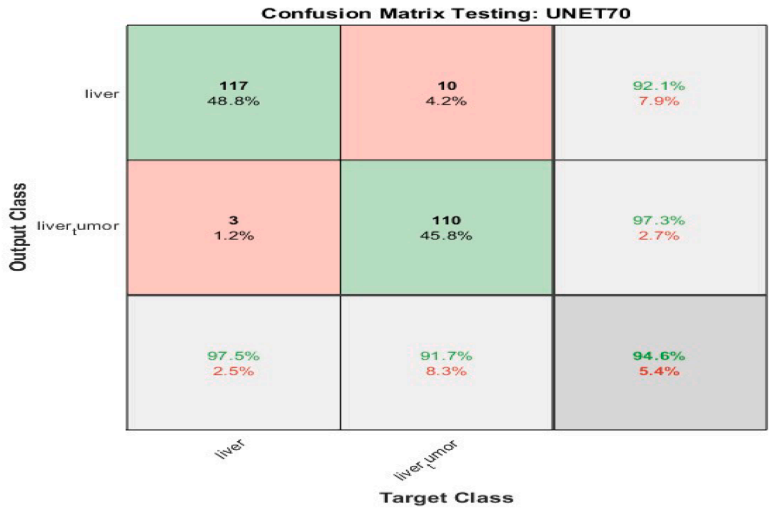

Fig. 7. Confusion matrix for testing.

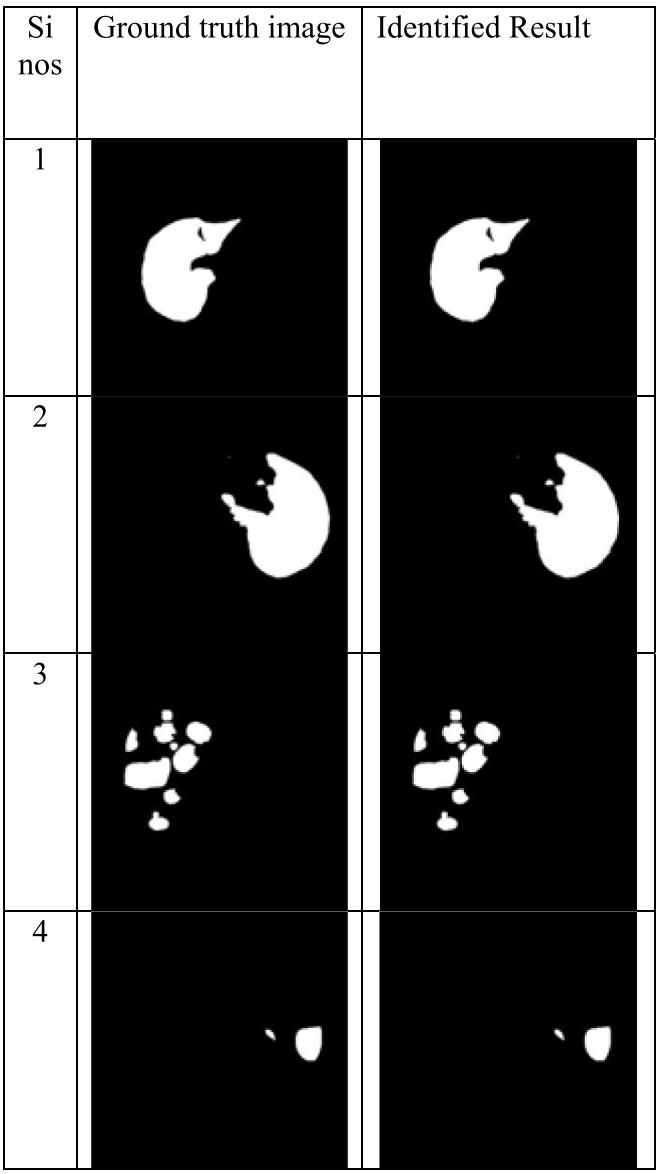

Fig. 8. Results of the proposed UNet70.

Table 5
Results of UNet70.

Parameters	Training Dataset (%)	Testing dataset (%)
Accuracy	92.82	94.58
Sensitivity	91.18	97.5
Specificity	94.47	91.66
Precision	94.28	92.12
Dice Score	92.70	94.73

Table 6
Different classifiers comparison.

Author	Accuracy (%)	Specificity (%)	Sensitivity (%)	Precision (%)	Dice Score (%)
MF [20]	78.50	70.40	86.60	77.40	81.80
SVM [21]	95.17	95.76	94.22	–	
KNN [21]	93.87	95.31	92.56	–	
NAÏVE BAYES [21]	91.94	93.65	89.91	–	
ANN [22]	88.89	82.88	95.23	–	
MLP [7]	89.15	88.01	90.38	89.05	
C4.5 [7]	95.02	96.12	93.89	94.41	
CNN [23]	88.00	–	–	–	
CNN [24]	89.58	–	–	–	
CNN [25]	86.90	–	–	–	
ENSEMBLE MODEL [26]	87.01	–	–	–	
GLCM [27]	89.00	–	–	–	
ALEXNET, VGG16 [28]	80.00	–	–	–	–
Proposed Method (UNet70)	94.58	91.66	97.5	92.12	94.73

Declarations

Ethics statement

The authors declare that the work described has not involved experimentation on humans or animals.

Ethics approval and consent to participate

We comply to the ethical standards. We give our consent to participate.

Consent for publication

All the authors are giving consent to publish

Availability of data and materials

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Funding

No funding from outside institute is availed.

Disclosure of potential conflicts of interest

The authors declare no conflict of interest, financial or otherwise.

Research involving human participation and/or animals

Not applicable.

Informed consent

Not applicable.

CRediT authorship contribution statement

Yashaswini Gowda N: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Methodology, Investigation, Data curation. **Manjunath R V:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation.

Declaration of competing interest

No conflict of interest to declare.

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